

Virtual homological torsion in low dimensions

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A mystery in 3-dimensions

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In hyperbolic 3-manifolds:

topological complexity of finite covers grows at a rate reflecting geometry



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Conjecture (Bergeron–Venkatesh)

Let M be a closed, hyperbolic 3-manifold and let

$$M = M_0 \leftarrow M_1 \leftarrow M_2 \leftarrow \cdots$$

be a cofinal tower of finite-sheeted normal covers of M , that is,

$$\bigcap_n \pi_1(M_n) = 1.$$

Then

$$\lim_{n \rightarrow \infty} \frac{\log(|\mathrm{Tor}(H_1(M_n; \mathbb{Z}))|)}{[\pi_1(M) : \pi_1(M_n)]} = \frac{\mathrm{vol}(M)}{6\pi}.$$



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- **Geometry** $\longleftrightarrow$ **Topology**: “Gauss-Bonnet on steroids” –

$$\left(\lim_{n \rightarrow \infty} \frac{\dim(H_1(\Sigma_n))}{\deg(\Sigma_n \twoheadrightarrow \Sigma)} \right) = \chi(\Sigma) = \frac{K \cdot \text{Area}(\Sigma)}{2\pi}$$

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- **Topology** $\longleftrightarrow$ **Analysis**: Prediction $\text{vol}(M)/6\pi$ obtained using analytic methods on the universal cover $\tilde{M} \cong \mathbb{H}^3$ of M (L^2 -torsion, Lück-Schick '99).

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- **Geometry** $\longleftrightarrow$ **Number theory**: **Roughly**, in arithmetic groups, homological torsion parametrizes field extensions $K/\mathbb{Q}$.

Open problem

Is there a finitely presented residually finite group G , and a residual normal chain

$$G \supseteq G_1 \supseteq G_2 \supseteq \cdots$$

(i.e. $\bigcap_n G_n = \{1\}$), such that

$$\lim_{n \rightarrow \infty} \frac{\log(|\mathrm{Tor}(G_n^{\mathrm{ab}})|)}{[G : G_n]} > 0 ?$$

Theorem (Sun '15)

Let M be a closed hyperbolic 3-manifold and let A be a finite abelian group. There exists a finite-sheeted cover $\widehat{M} \rightarrow M$ such that

$$H_1(\widehat{M}; \mathbb{Z}) = A \oplus \cdots .$$

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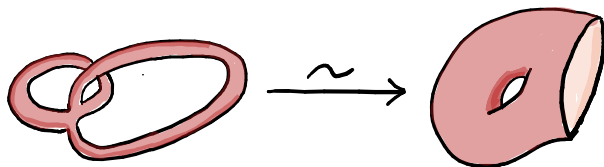
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“Abundance of virtual homological torsion”

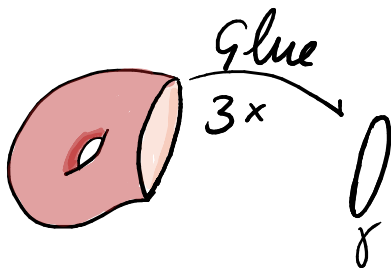
Similar proof works for all 3-manifolds with empty or toroidal boundary which are not graph manifolds (Chu–Groves '22).

Virtual homological torsion in 3-manifolds: a recipe

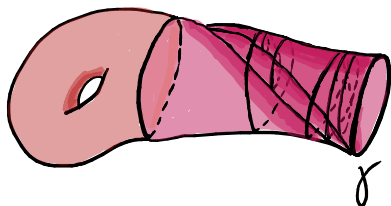
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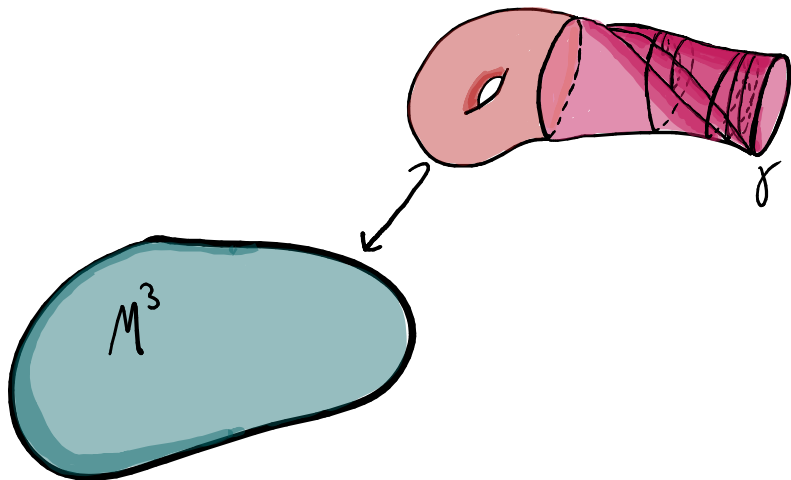
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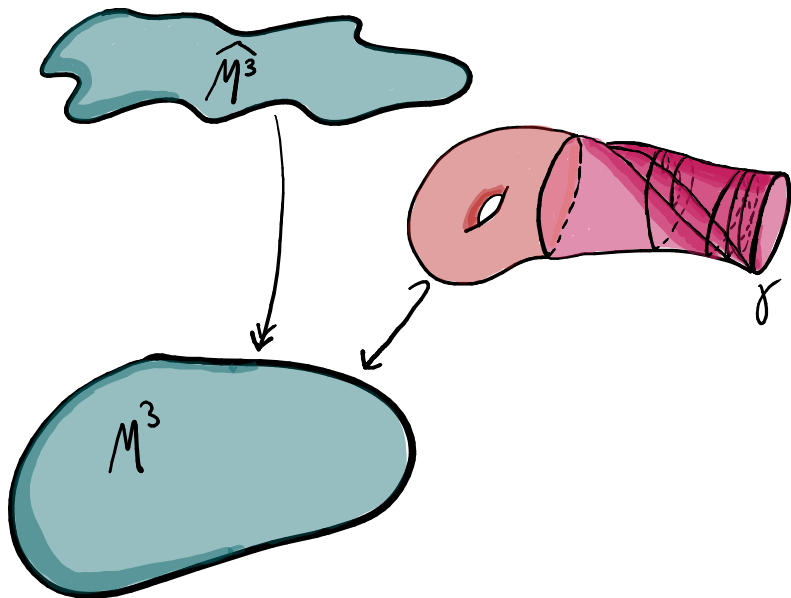
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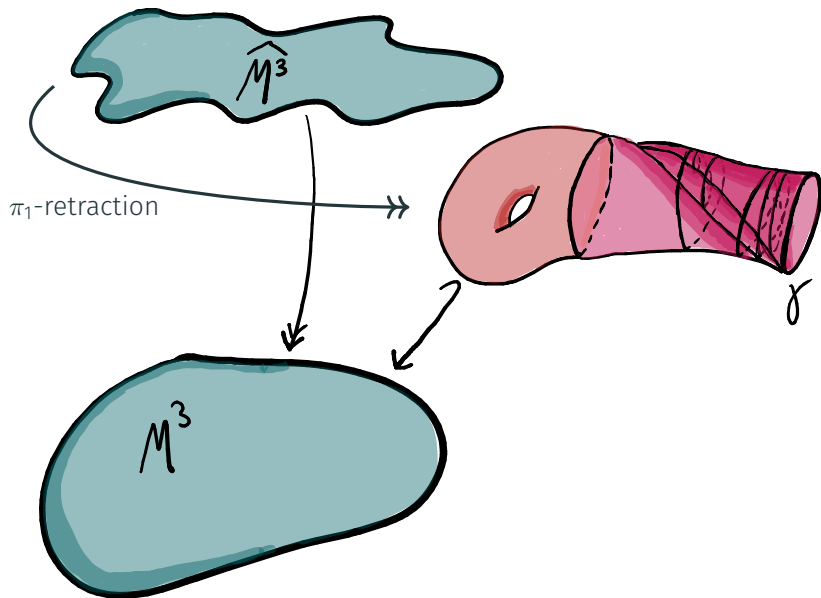
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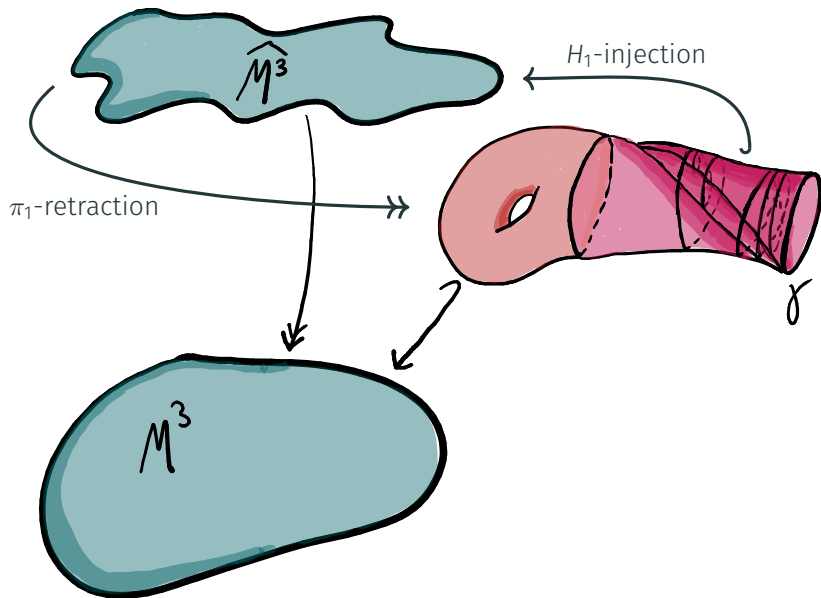
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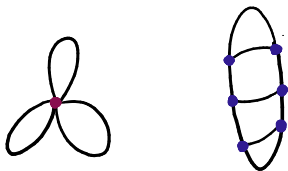


(Hyperbolic) Graphs of free groups with cyclic edges

(Fundamental groups of) graphs joined together by cylinders.

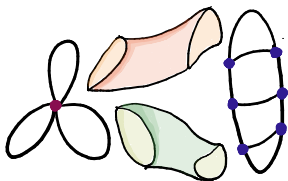
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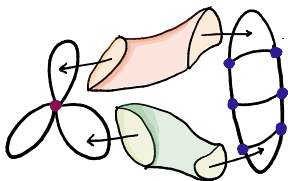
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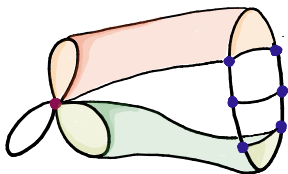
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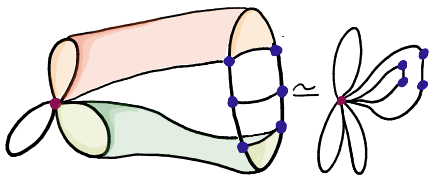
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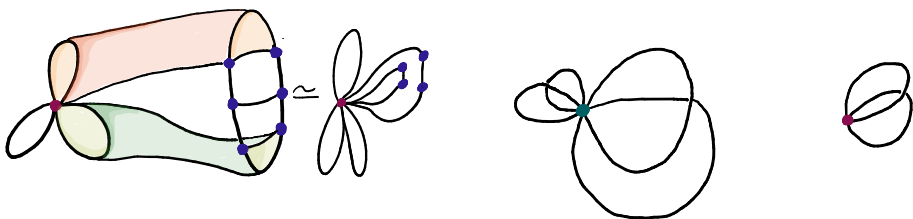
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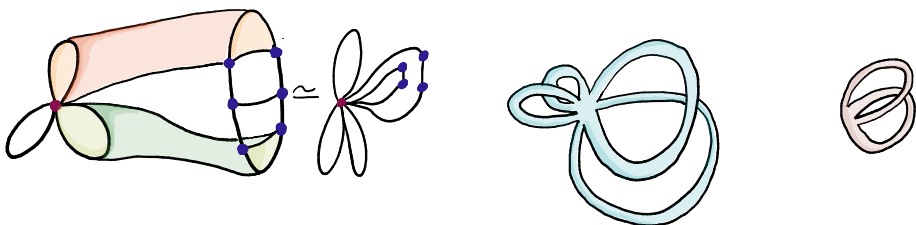
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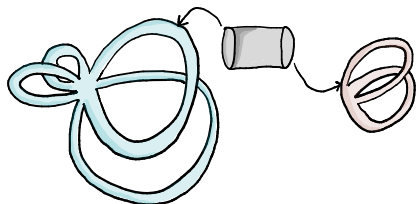
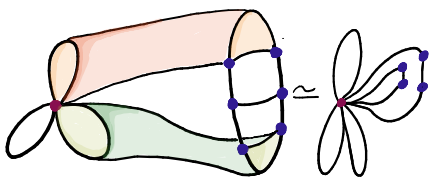
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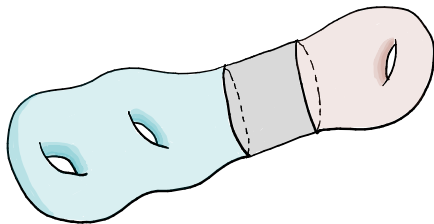
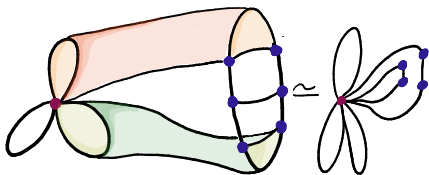
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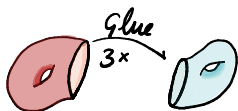
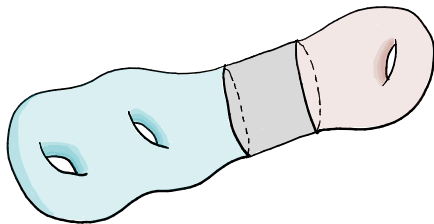
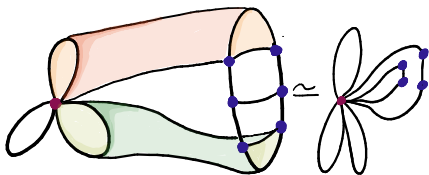
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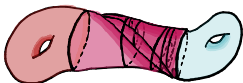
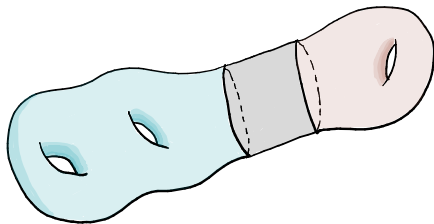
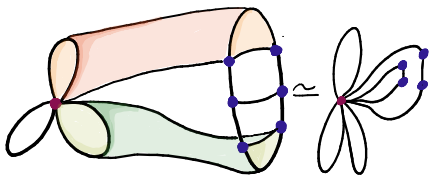
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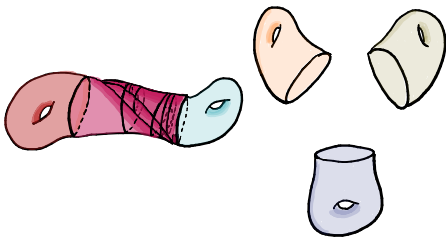
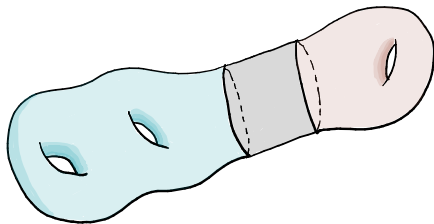
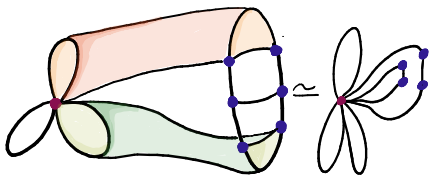
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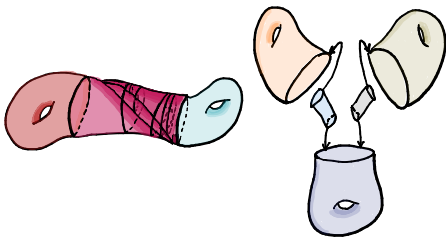
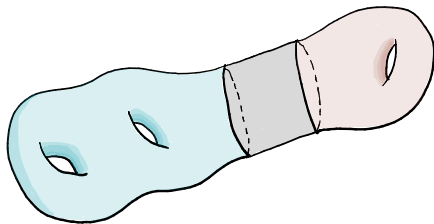
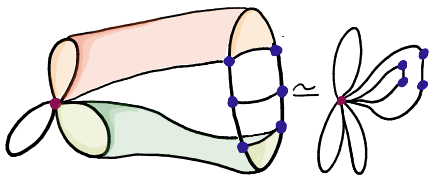
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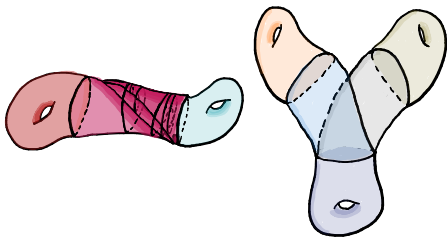
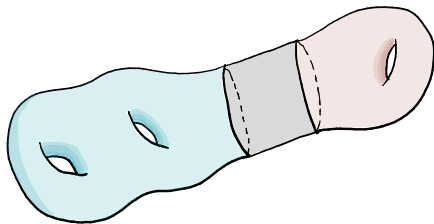
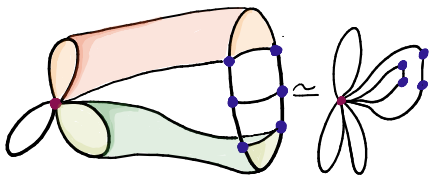
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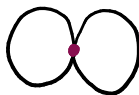
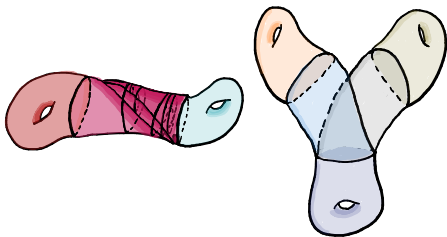
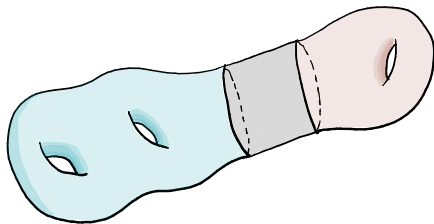
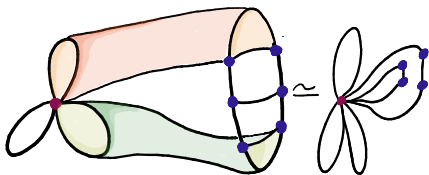
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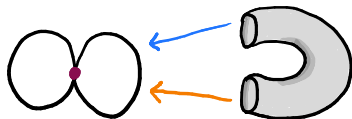
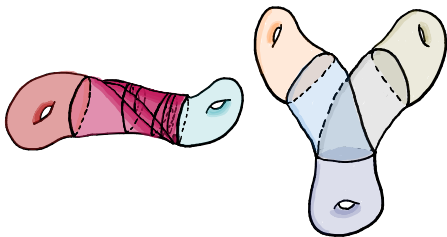
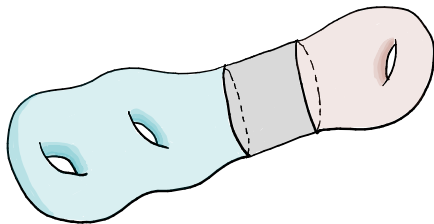
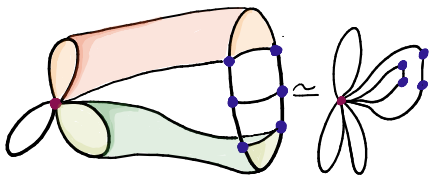
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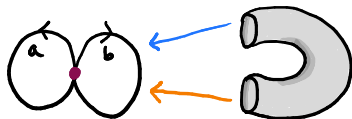
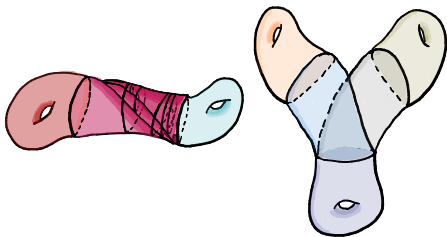
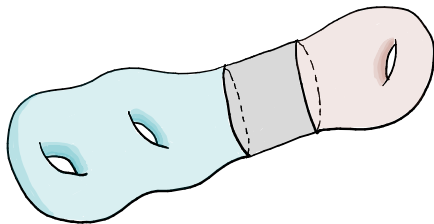
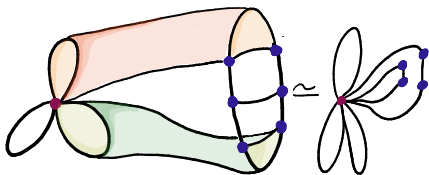
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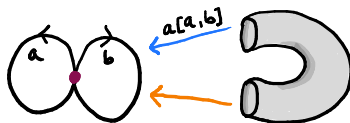
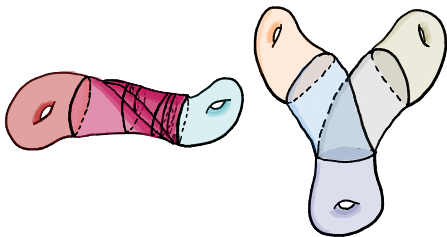
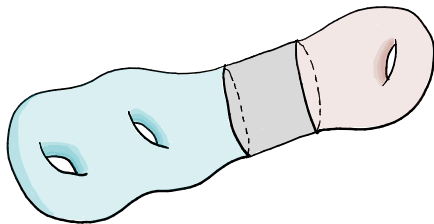
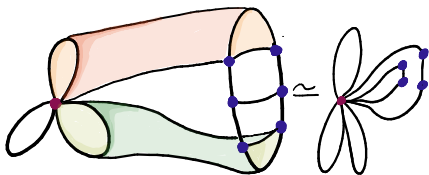
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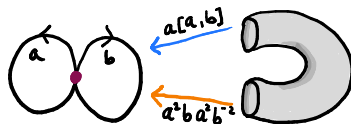
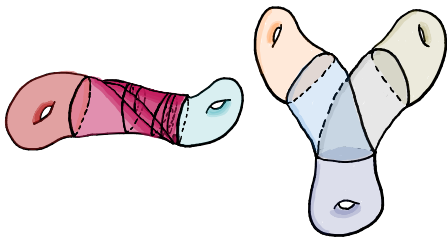
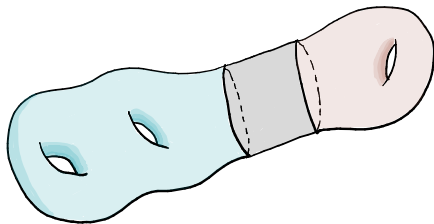
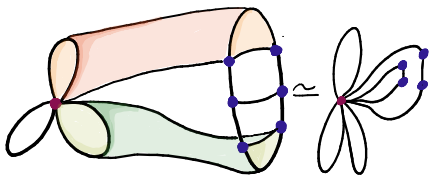
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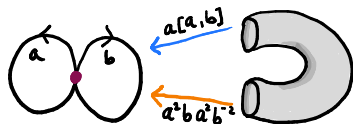
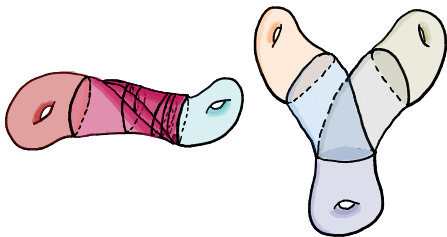
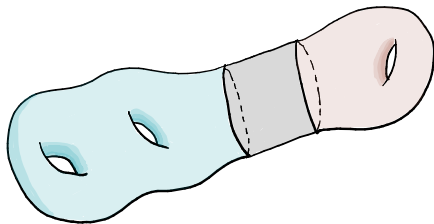
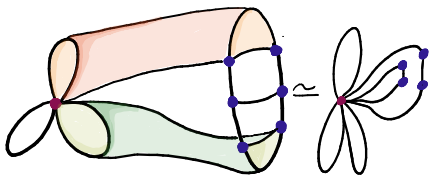
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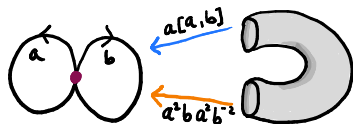
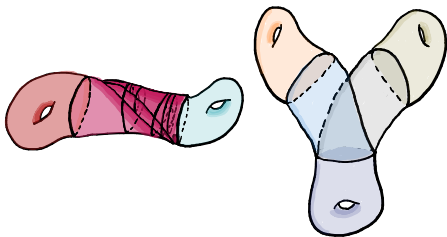
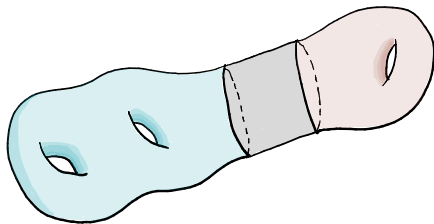
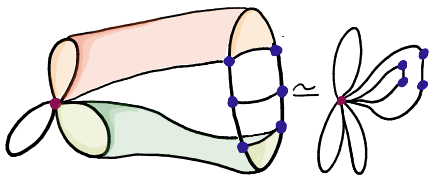
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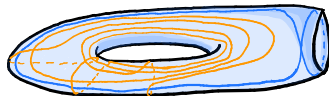
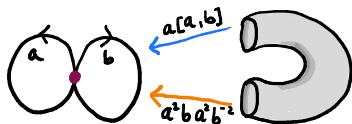
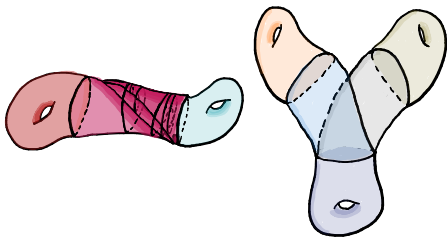
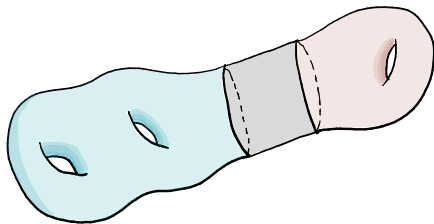
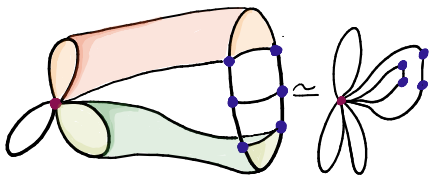
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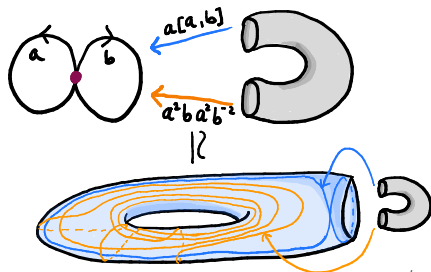
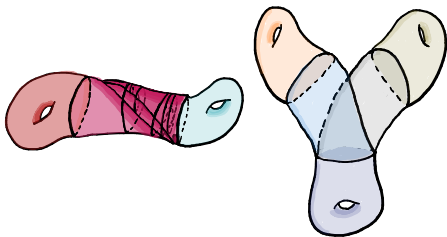
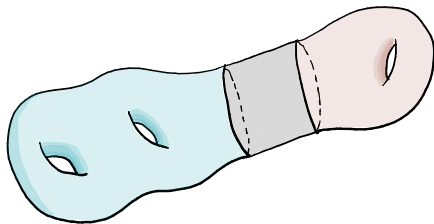
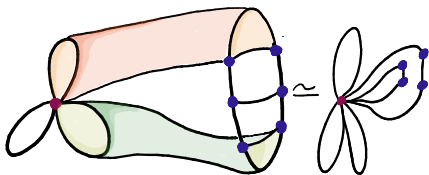
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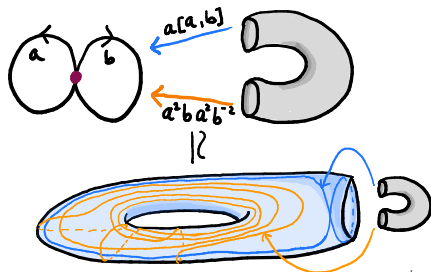
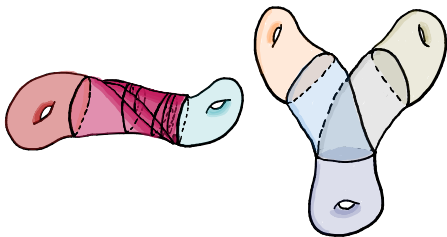
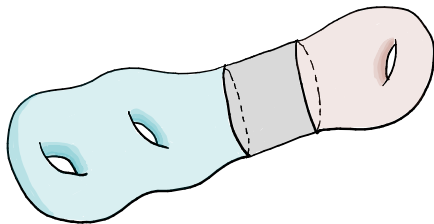
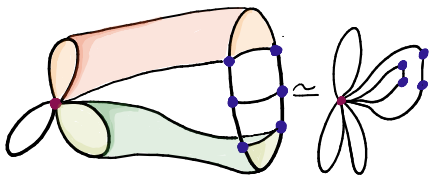
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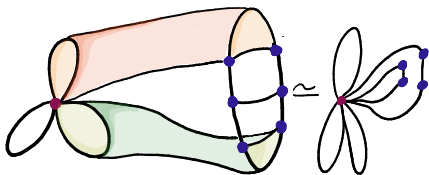
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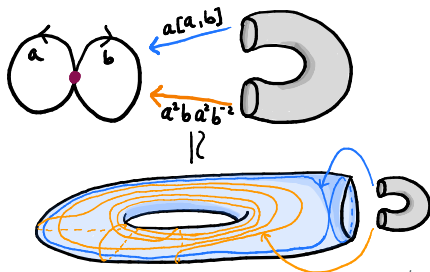
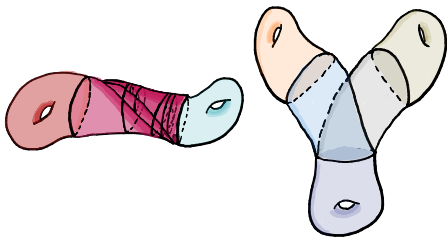
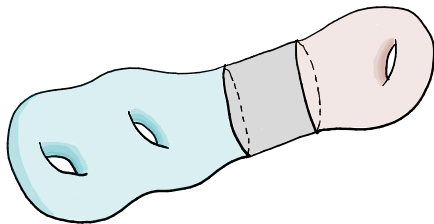


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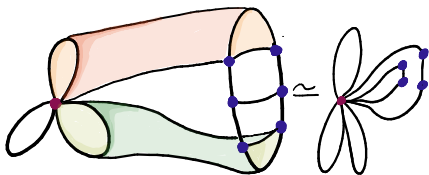


Free and surface groups
(and their free products)

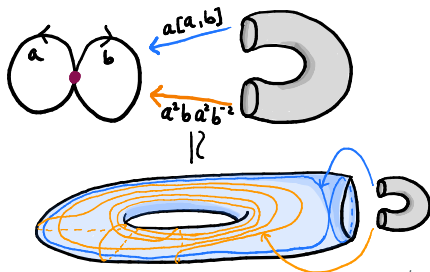
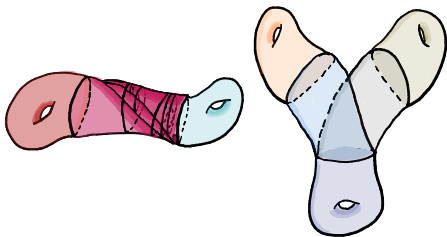
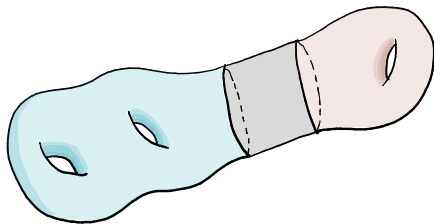


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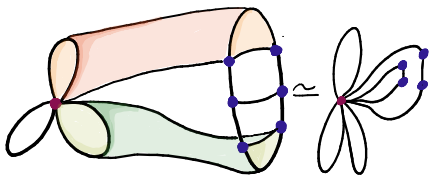


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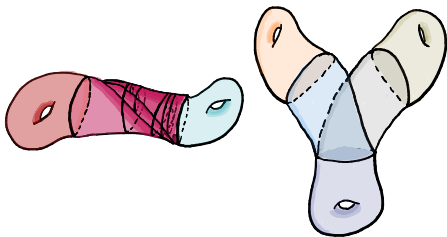
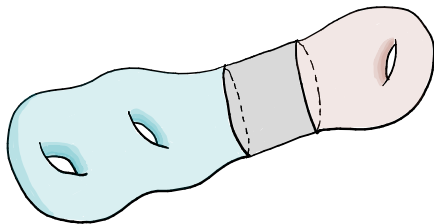


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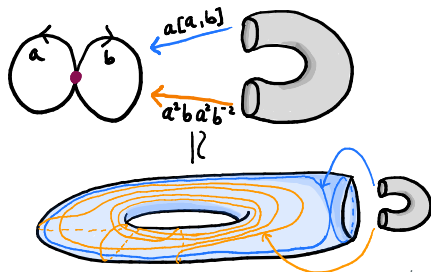
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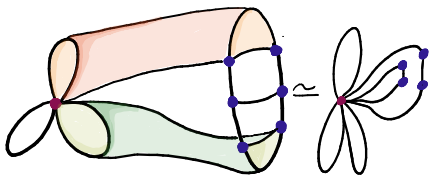


"More than a surface"

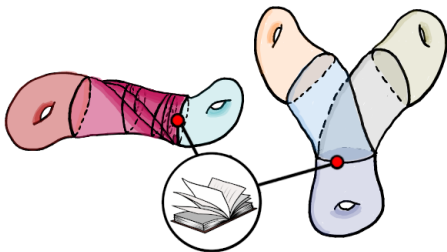
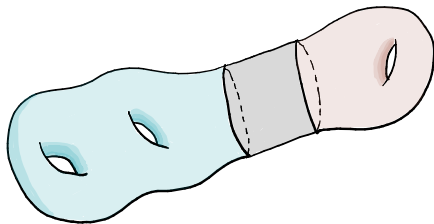


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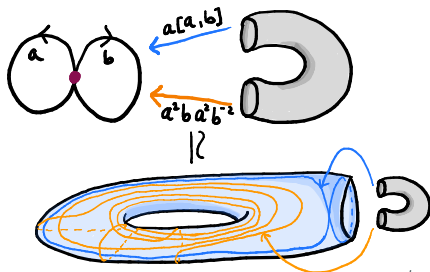
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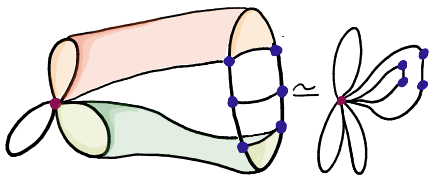


"Books of I -bundles"

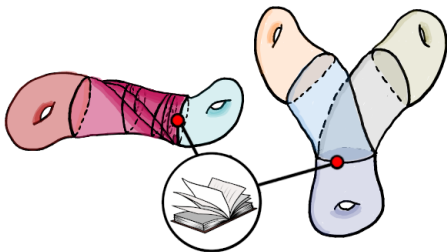
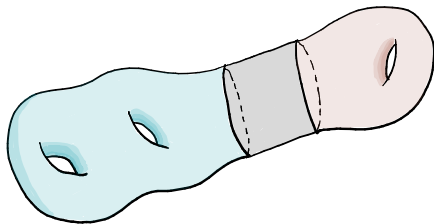


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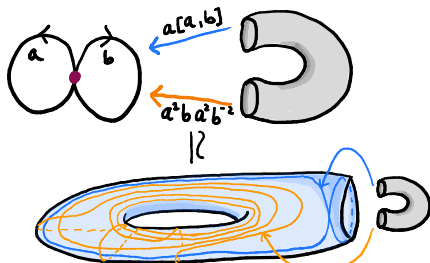
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Free and surface groups
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"Books of l -bundles"



"Has a *rigid* vertex"

Homological torsion: Existence

Homological torsion: Existence

Theorem (Ascari-F '25)

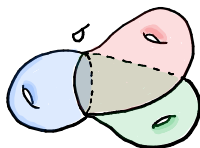
Let G be a hyperbolic graph of free groups with $\mathbb{Z}$ edges, which is not a free product of free and surface groups. Then G has abundant virtual homological torsion, that is,

for every finite abelian group A , there is a finite-index subgroup $G_0 \leq G$ such that

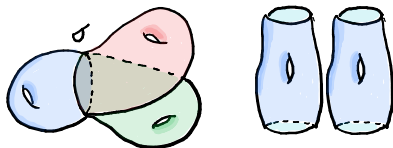
$$G_0^{\text{ab}} = A \oplus \cdots .$$

Homological torsion: Systems of surfaces

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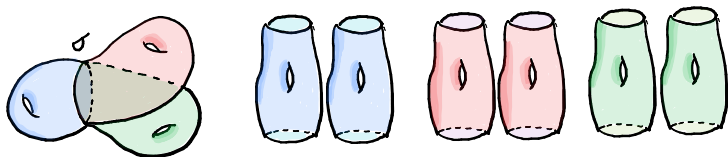
Homological torsion: Systems of surfaces



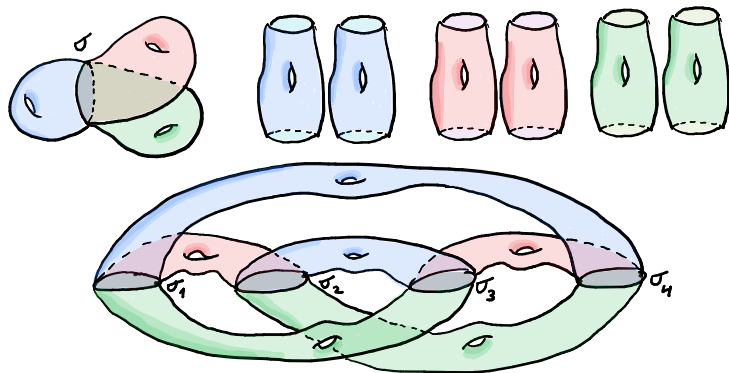
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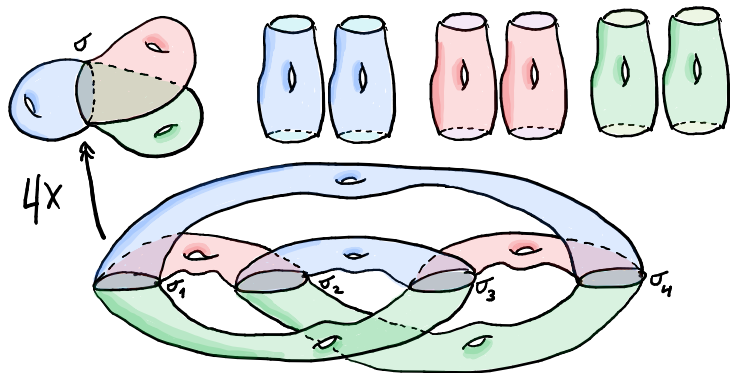
Homological torsion: Systems of surfaces



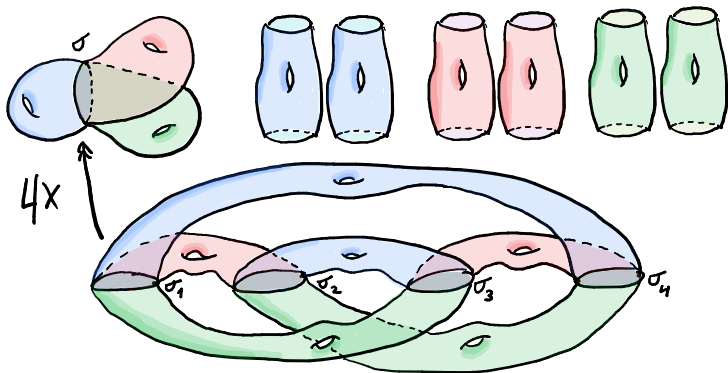
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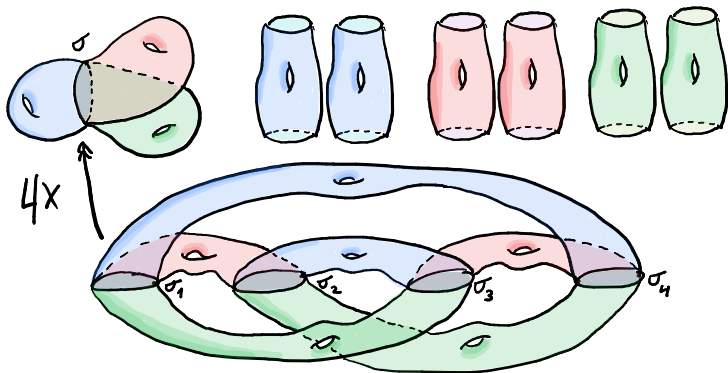


Homological torsion: Systems of surfaces



In first homology:

Homological torsion: Systems of surfaces



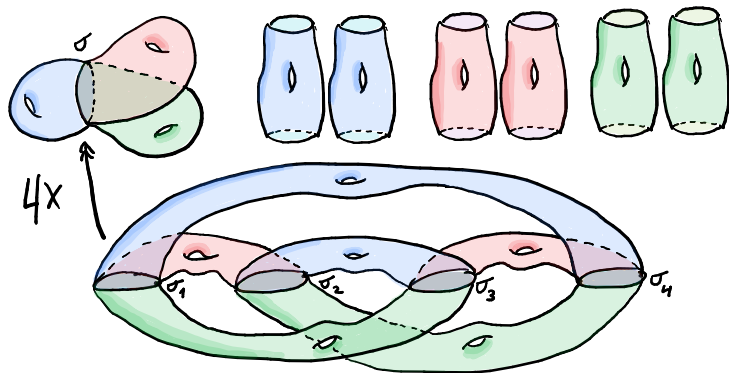
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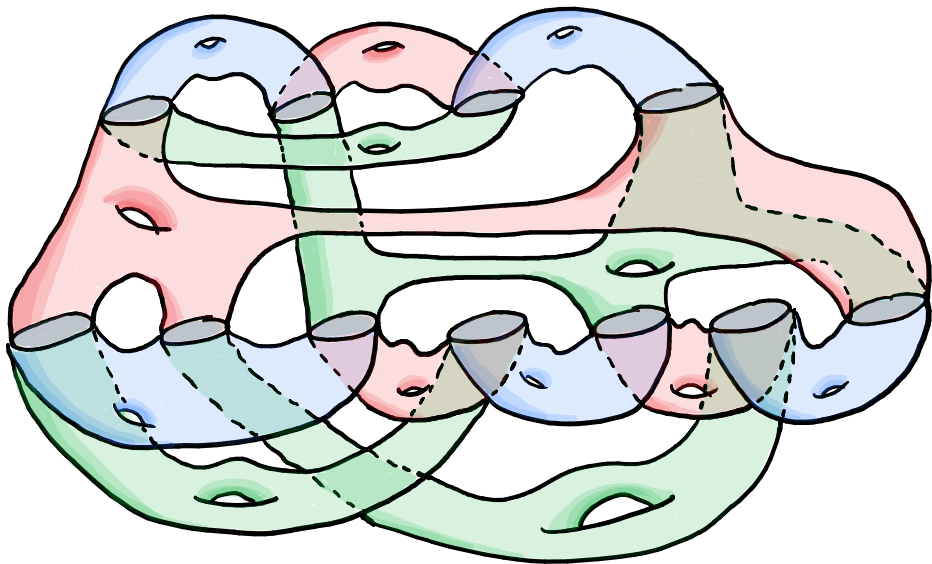
$\Rightarrow$

$$\sigma_1 = \sigma_2 = \sigma_3 = \sigma_4$$

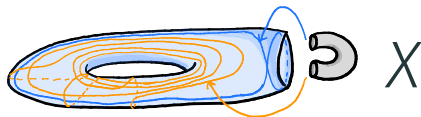
$$2 \cdot \sigma_1 = 0$$

Homological torsion: Two and three

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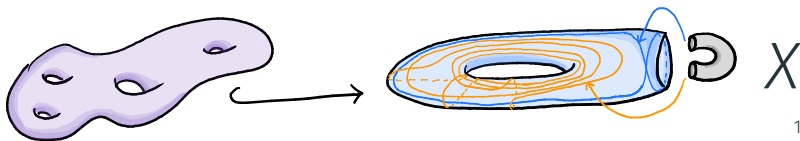
Enter a rigid vertex



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Theorem (Wilton '18)

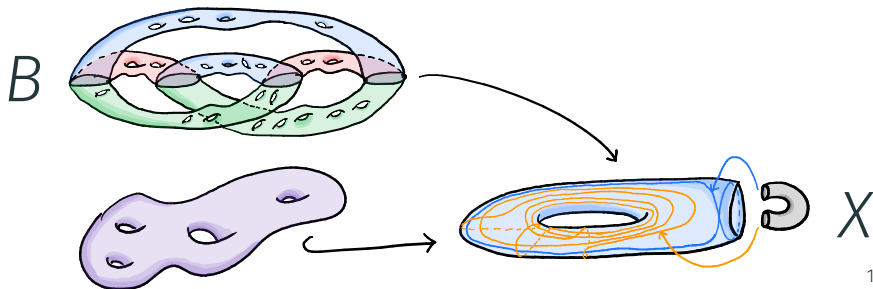
If G is not free then G has a surface subgroup.



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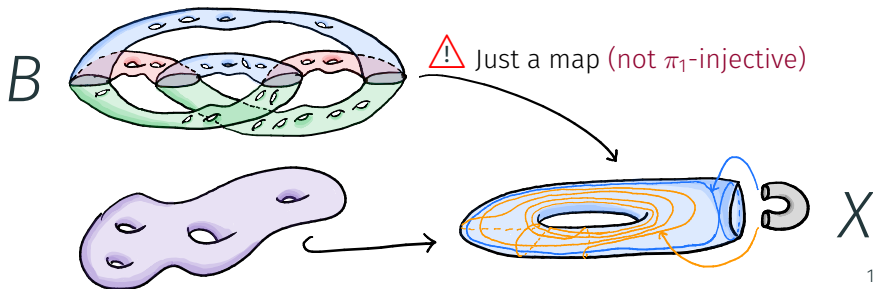
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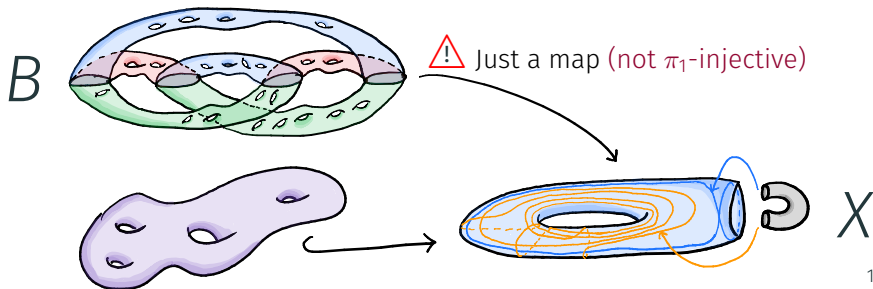
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Let F be a free group and let $g \in [F, F]$. Recall $\text{cl}(g) = \min\{\ell \mid g = \text{product of } \ell \text{ commutators}\}$.



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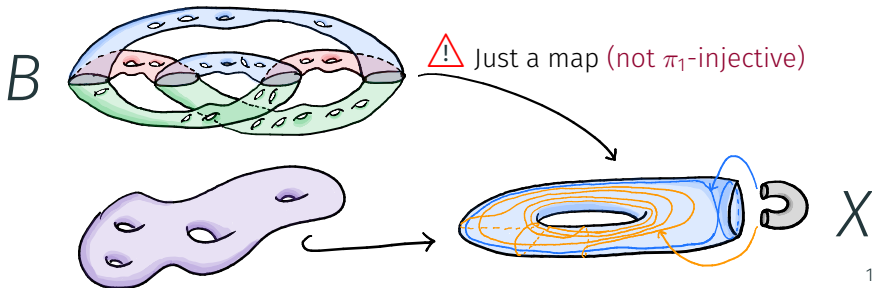
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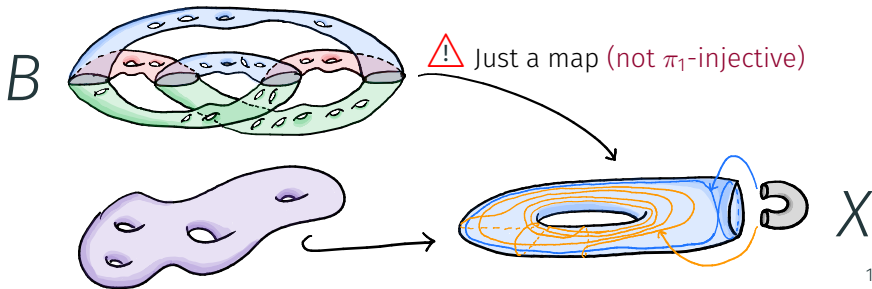
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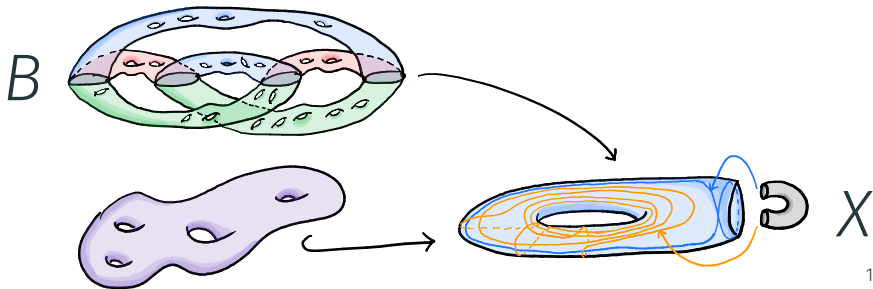
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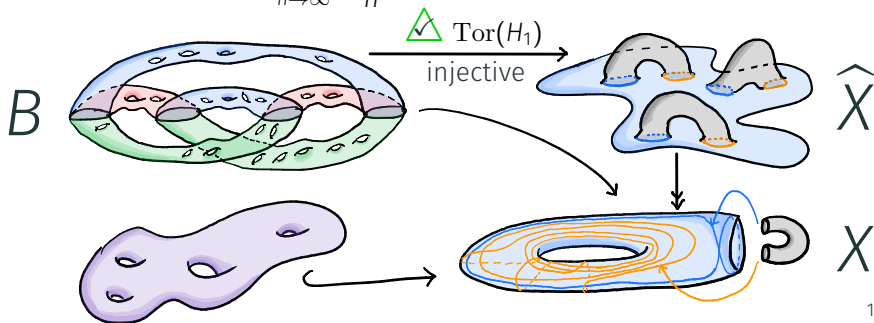
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Let it grow

Theorem (F '25, F '26)

In graphs of free groups with $\mathbb{Z}$ edges, homological torsion grows subexponentially along any exhausting tower of finite-index normal subgroups.

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- **Tension:** These can have abundant virtual homological torsion, which can only grow “slowly”.

Let it grow

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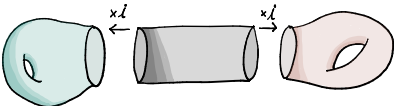
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- Bergeron-Venkatesh: such behaviour expected in dimension ≥ 4 .

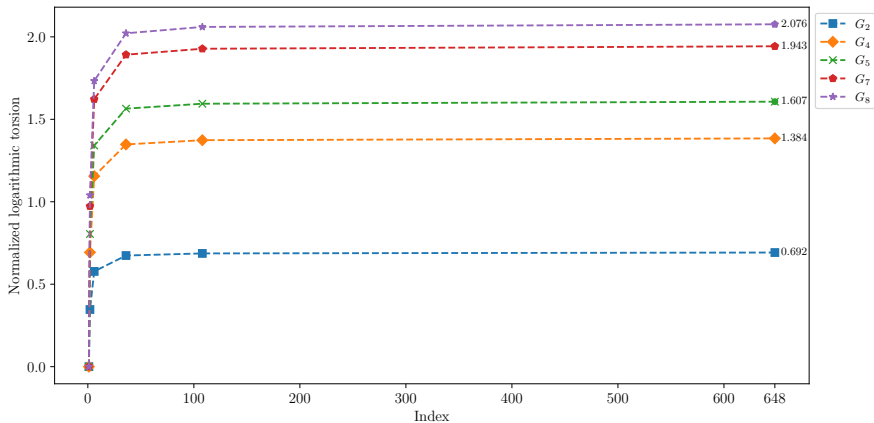
A computation: unwrapping cylinders

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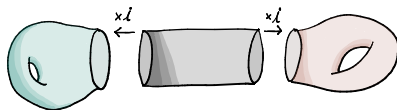
$$G_i = \langle a, b \rangle_{[a,b]^i=[c,d]^i} \langle c, d \rangle$$


The diagram illustrates the unwrapping of cylinders. It shows three cylinders arranged horizontally. The leftmost cylinder is teal and has a small loop on its side. The middle cylinder is gray and is the narrowest. The rightmost cylinder is pink and also has a small loop on its side. Two arrows, both labeled x_i , point from the gray cylinder towards the teal and pink cylinders respectively, indicating a mapping or transformation.

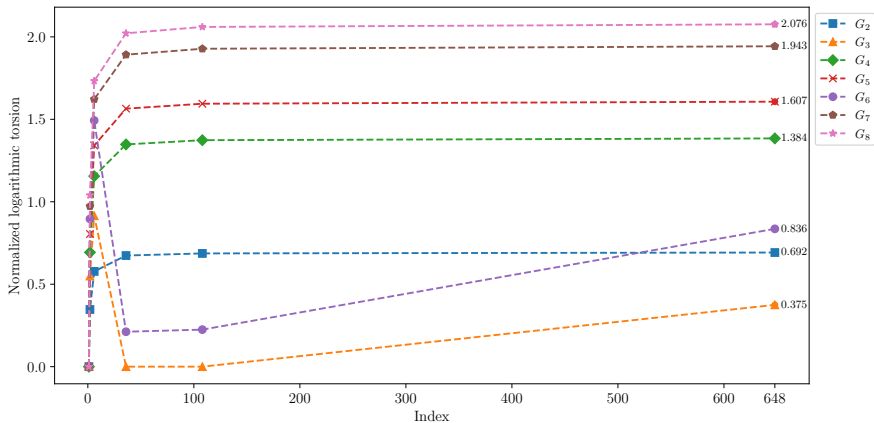
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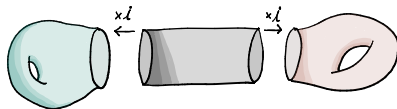
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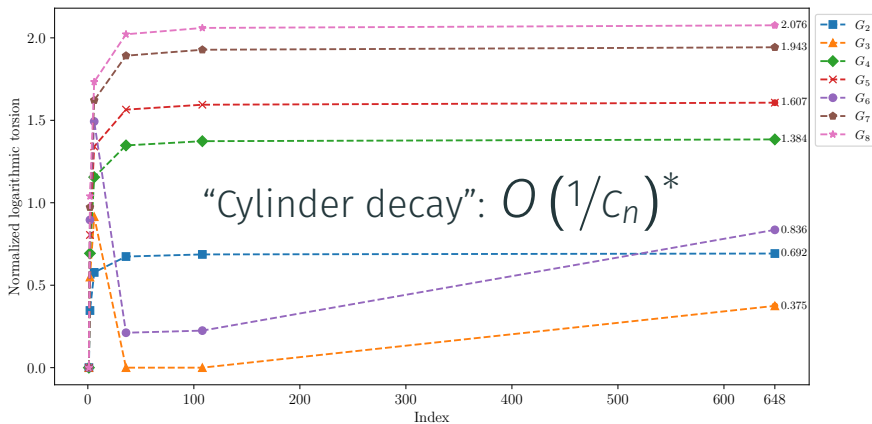
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A computation: unwrapping cylinders



* C_n measures the wrapping degree of the cylinder in the n^{th} cover.

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The landscape

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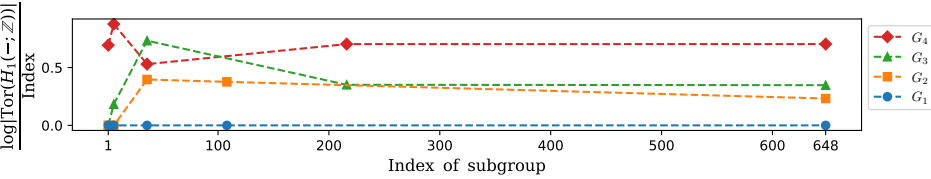
$\text{Tor}(H_1)$	(Hyp) 3-manifolds	Surface groups	Books of I -bundles	Rigid vertex
Abundance	 (Sun '15)		 (Ascari-F '25)	 (Ascari-F '25)
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$$G_1 = \pi_1 \left(\text{Diagram of a genus-1 surface} \right)$$

G_{i+1} obtained from G_i by adding a single letter to its presentation

Thank you for listening!

